

system resources via an Application Programming Interface. By contrast, a server which is running three containerised applications, as with Docker, runs just a single operating system, and each of the containers shares that operating system kernel with the other containers. We need to note that the shared parts of the operating system are read only, while each of the containers has its own way to access the operating system for writing. This means that containers are much more lightweight and use far fewer resources than virtual machines. A container may be only a few megabytes in size, whereas a virtual machine with its own entire operating system may be several gigabytes in size. This is why a single server can host far more containers than virtual machines.

1. Containers are easily packaged and designed to run anywhere. There can be multiple containers deployed in a single VM.
2. Also, it may take several minutes for virtual machines to boot up their operating systems and then begin running the applications they host, whereas containerised applications can be started almost instantly.

Docker: An open source software containerisation platform

Docker is an open source software containerisation platform designed to make it easier to create, deploy and run various applications by using containers. According to the official Web page of Docker, its containers wrap up a piece of software into a complete file system that contains everything it needs to run: runtime, code, system tools or system libraries – anything we can install on a server. This ensures that it will always run the same, irrespective of the environment it is running in.

Docker provides an additional layer of abstraction and even automation of virtualisation at the operating system level on Linux. It actually uses different resource isolation features of the Linux kernel, like cgroups and kernel namespaces, and also a union-capable file system such as OverlayFS to allow independent ‘containers’ to run within a single Linux instance. This helps in avoiding the overhead of starting and then maintaining virtual machines, which significantly boosts its performance and also reduces the size of the application. The support of the Linux kernel for namespaces isolates an application’s view of the operating environment—including process trees, user IDs, network and mounted file systems—while the kernel’s cgroups helps in limiting resources, including the memory, CPU, block I/O and network.

As various actions are done to a Docker base image, different union file system layers are created as well as documented, such that each layer fully describes the way to recreate an action. This strategy enables Docker’s lightweight images, since only layer updates need to be propagated in this. It implements a high-level API to provide lightweight

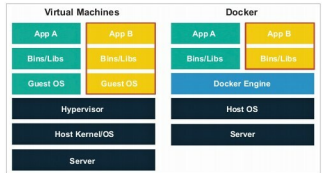


Figure 1: Architecture of virtual machines vs Docker (Image credits: Google images)

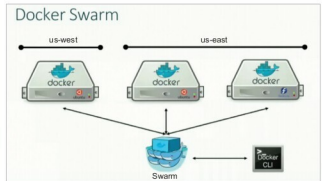


Figure 2: A Docker swarm (Image credits: Google images)

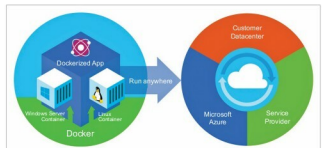


Figure 3: Microsoft Azure support for Docker (Image credits: Google images)

containers that run different processes in isolation. Docker helps enable portability and flexibility wherever the application can run – whether on-premise, the private cloud, the public cloud, bare metal, etc. Docker accesses the different virtualisation features of the Linux kernel either directly, using the libcontainer library available in Docker 0.9, or indirectly via libvirt and LXC (Linux Containers). Since Docker containers are very lightweight, even a single server or a virtual machine can run several containers simultaneously. According to an analysis done in 2016, a typical Docker use case involves running five containers per host, but if we ponder over this, then that many organisations can actually run 10 containers or even more.

Using Docker to create and manage containers may actually simplify the creation of highly distributed systems by allowing multiple worker tasks, applications and other processes to run autonomously on a single physical machine